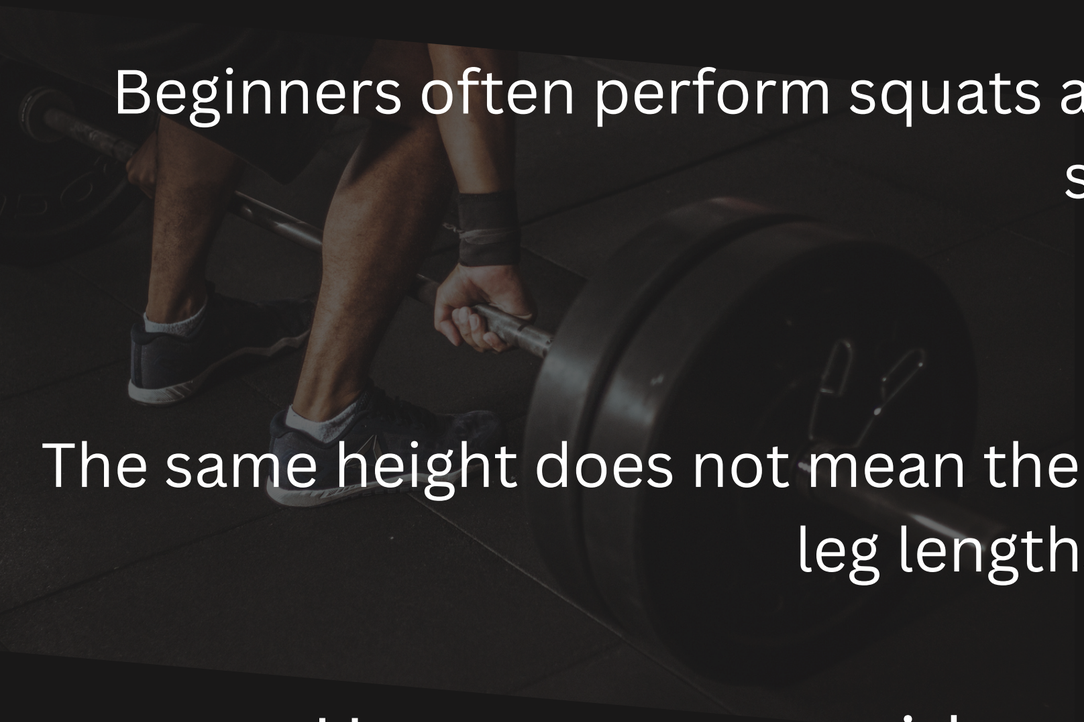
Problem statement



Beginners often perform squats and deadlifts without continous professional supervision.

The same height does not mean the same body proportions — torso, thigh, and lower-leg lengths can differ significantly

How can we provide real-time exercise-form feedback that considers an individual’s body proportions instead of applying the same fixed movement criteria to everyone ?

Proposed solution

Our Solution: FormGuard AI

A computer-vision-based fitness assistant that analyzes squat and deadlift movements through a normal camera.

📷 CAMERA

↓

POSE DETECTION

↓

BODY LANDMARKS

↓

BODY PROPORTION PROFILE

↓

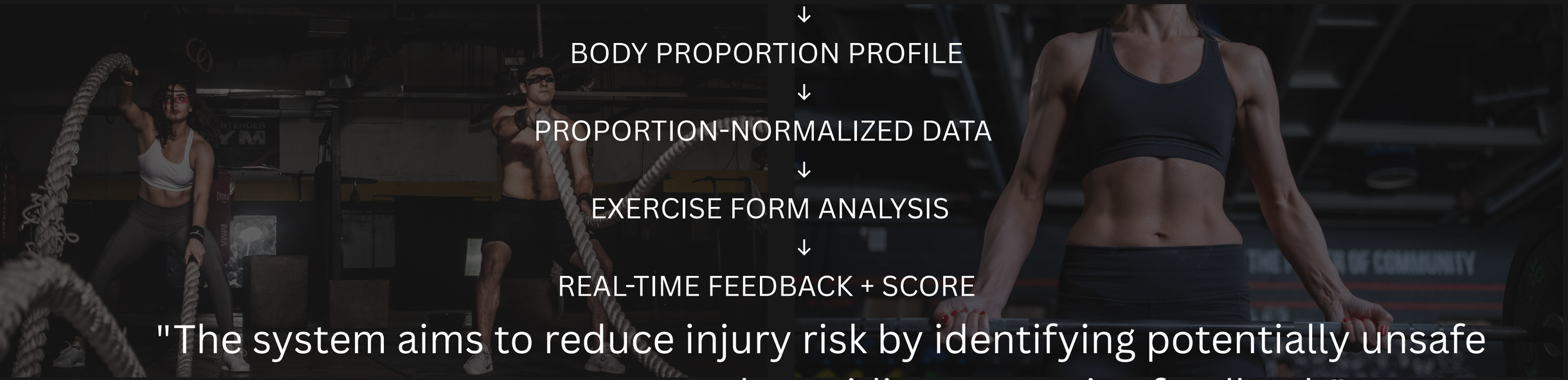
PROPORTION-NORMALIZED DATA

↓

EXERCISE FORM ANALYSIS

↓

REAL-TIME FEEDBACK + SCORE



"The system aims to reduce injury risk by identifying potentially unsafe movement patterns and providing corrective feedback."

Target users/ Imapct

Target Users

Gym Beginners – Learn correct squat & deadlift form

Home Workout Users – Get real-time guidance without a trainer

Fitness Trainers – Additional tool for form monitoring

Expected Impact

Challenge

- Beginners don't know if their form is correct
  Real-time form feedback
- One standard doesn't fit every body
- No trainer available continuously

formGuard AI’s contribution

- Real-time form feedback
- Body-proportion-aware analysis
- Camera-based assistance

Personalized Body-proportion Analysis

Before analyzing the exercise, FormGuard AI estimates the user's relative body proportions.

Important landmarks

Torso

Shoulder

●

|

|

●

Hip

Leg

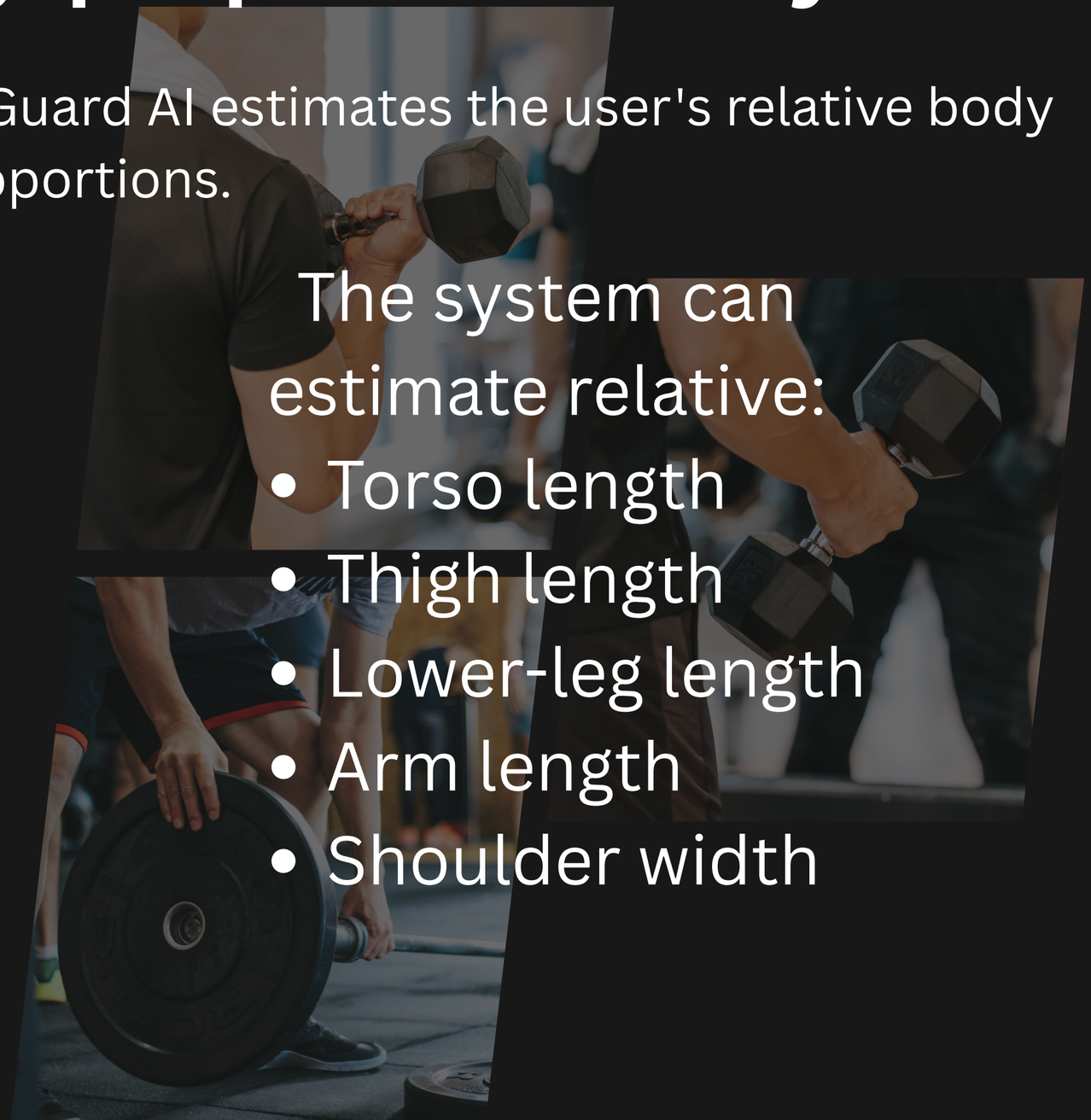
Hip

●

●Knee

|

● Ankle



The system can estimate relative:

- Torso length
- Thigh length
- Lower-leg length
- Arm length
- Shoulder width

Implementation

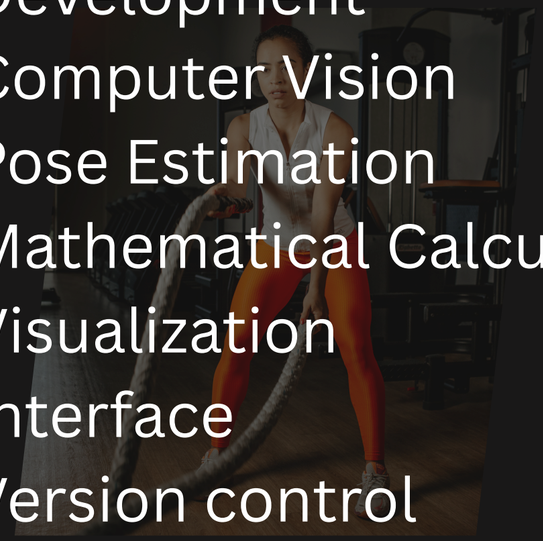
How we will Build it

1. Camera Input
Webcam captures the exercise in real time.
↓
2. Pose Detection
MediaPipe detects body landmarks: Shoulder → Hip → Knee → Ankle
↓
3. Personal Calibration
Neutral standing position is used to estimate the user’s body proportions.
↓
4. Mathematical Analysis
Python calculates joint angles, relative distances, body proportions & movement trajectory.
↓
5. Exercise Analysis
Squat: Hip movement • Knee angle/alignment • Torso position • Depth & control
Deadlift: Hip movement • Knee position • Torso position • Shoulder–hip relationship • Movement consistency
↓
6. Real-Time Feedback
Form Status + Specific Correction + Form Score

Technology stack

Purpose

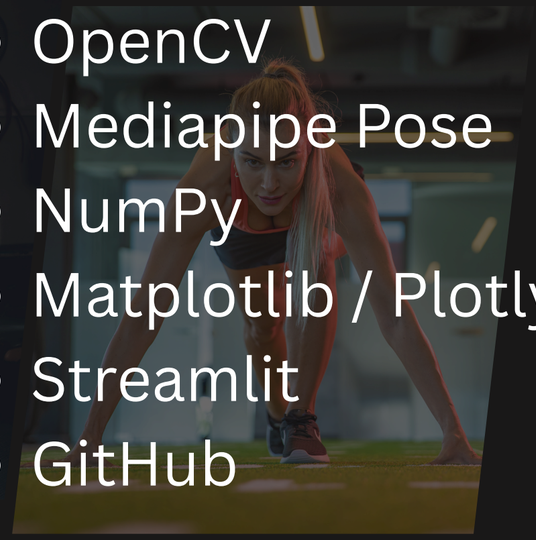
- programming
- Development
- Computer Vision
- Pose Estimation
- Mathematical Calcuation
- Visualization
- Interface
- Version control





Technolgy

- Python
- Pycharm
- OpenCV
- Mediapipe Pose
- NumPy
- Matplotlib / Plotly
- Streamlit
- GitHub

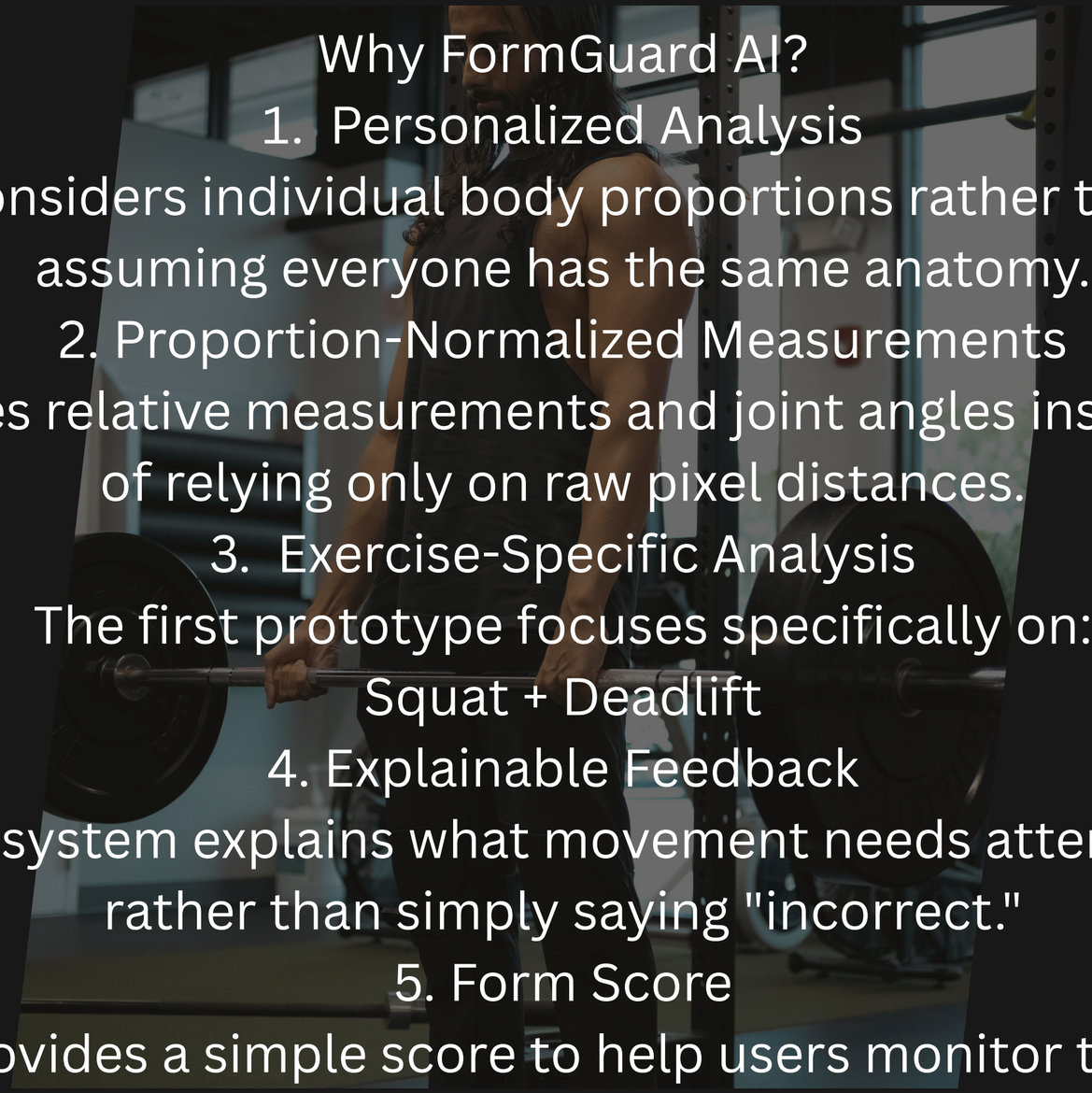




Core Technology flow

Python → OpenCV → MediaPipe Pose → Body Landmarks → NumPy Calculations → Form Analysis → Feedback

Innovation/ Uniqueness Future scope



Why FormGuard AI?

1. Personalized Analysis

Considers individual body proportions rather than assuming everyone has the same anatomy.

2. Proportion-Normalized Measurements

Uses relative measurements and joint angles instead of relying only on raw pixel distances.

3. Exercise-Specific Analysis

The first prototype focuses specifically on:

Squat + Deadlift

4. Explainable Feedback

The system explains what movement needs attention, rather than simply saying "incorrect."

5. Form Score

Provides a simple score to help users monitor their technique over time.

Future Scope

After successfully developing the squat + deadlift system:

Phase 2

→ Bicep curls

→ Lunges

→ Shoulder exercises

→ Push-ups

Phase 3

→ More exercises

→ Personalized workout analysis

→ Progress tracking

→ Mobile/web deployment

→ More advanced ML-based movement analysis